

How We Get Where We Are

People, Institutions, and Luck in an Unequal World

Honggao Cao

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Preface

This manuscript grew out of an interest in inequality and the explanations we give for it. People differ in ability, effort, interests, and judgment. Yet recognizing those differences leaves much to understand about the rewards they receive, the setbacks they face, and the opportunities they can pass to others. I wanted to follow how differences become consequential, and why some consequences reach so much further than the activity that first produced them.

The question matters because an account of how someone reached a position shapes how we understand the person and what we think can be changed. A result read only as a measure of talent or effort suggests one explanation. Looking also at access, ownership, obligations, and timing may reveal something else that needs attention. The intention here is to keep individual action in view while examining the circumstances through which it becomes effective.

Following that concern led beyond the distribution of income or wealth. An opportunity must be usable; a service must be provided; a cooperative arrangement must be sustained. Understanding who benefits therefore also means asking how people build productive capacity, respond to interruptions, and alter the rules under which they act. Inequality remains the starting concern, but the inquiry also includes what people can accomplish together and how an unfavorable path might be changed.

The discussion is written for readers willing to look closely at familiar experiences without first becoming specialists in each subject. The cases range across housing, farming, health care, technology, finance, and irrigation. Their value here lies in helping us see connections that are easy to miss when a result is viewed alone. The research supplies evidence and challenges; the examples give us something concrete to follow.

My hope is that the reader comes away better able to ask what made an achievement possible, where an obstacle arose, and what an earlier outcome leaves available for the next decision. We need those explanations both to recognize genuine contributions and to judge which arrangements and possibilities deserve to be preserved or changed.

San Francisco, September 2026

Introduction

Imagine two people opening small businesses with comparable skills, experience, and savings. Both work hard to attract customers. One secures an early contract; the other narrowly misses it. The contract pays for better equipment and provides a reference for the next customer. The other owner spends the same months seeking work while drawing down savings.

Several years later, one business has expanded while the other has closed. The owners' decisions belong in the explanation. So do the opportunities available when those decisions were made. An initial difference in earnings has become a difference in equipment, customer relationships, and the ability to survive a difficult period. The later gap raises a question:

Why can relatively modest differences in individual capabilities coexist with much larger differences in rewards, resources, and influence?

The wealth-distribution literature documents pronounced concentration at the top across countries and periods. [1] Explaining who obtained the first contract is different from explaining its value or how long its consequences lasted. An accurate ranking of performance still leaves the size of the reward difference to be explained. In our example, we must also examine the customer's selection, the opportunity to expand, and the arrangements allowing one gain to support another.

In a Dutch research-funding program, applicants just above an award threshold accumulated, on average, more than twice as much subsequent funding over eight years as applicants just below it, excluding the initial grant. Part of the difference arose because unsuccessful applicants entered later competitions less often. The initial allocation thus affected participation in later competitions as well as the money already received. [2]

To understand how outcomes are produced, begin with the conditions operating at a particular time. What can people do, what resources and relationships can they use, and how do rules and organizations translate those positions into benefits and burdens? This is the point-in-time view.

Alongside skills, we need to know who has the equipment, time, finance, and access needed to use them. We also need to know which advantages occur together. In the business example, equipment and skill must be brought together, and their productive result must reach a customer. The combinations and connections matter alongside the components.

The process through which people make choices can matter too. In an experimental music market, Salganik, Dodds, and Watts gave participants access to the same set of unfamiliar songs while varying information about previous participants' choices. Stronger social influence increased both inequality and unpredictability of success. The songs' appeal still mattered, but it did not uniquely determine their success. [3]

The dynamic view follows how these present combinations developed. The successful business's equipment and customer record are both current resources and consequences of earlier activity. Today's outcome becomes one of tomorrow's conditions; a later snapshot records what these transitions have produced.

The two views belong together. An outcome distribution reflects how the components operate together at a particular time, while the dynamic account follows how those components, their combinations, and the arrangements governing them developed. The resulting distribution need not resemble the distribution of any single characteristic entering the process.

Property makes the transition concrete. In our business example, an asset bought with earlier earnings can remain available to its owner. Whether the owner can use it, receive its returns, borrow against it, or transfer it to someone else depends on the rights attached to it. These possibilities carry an earlier result forward in different ways. Someone receiving the asset later could begin with a resource position they did not create. The relevant history extends beyond the present holder's own activity.

Individuality, institutions, and luck meet within such histories. The owners' skills, effort, and choices remain consequential. Institutions shape what they can claim and retain. The timing of a customer's arrival or an unexpected interruption may lie outside an owner's control. That timing can be luck from the owner's perspective while arising from decisions made elsewhere.

Their interaction can also moderate or reverse a difference. Shared equipment, another accessible contract, or temporary support might allow the second business to continue. An early advantage might disappear. The history includes what sustains a difference and what makes recovery possible.

The inquiry therefore reaches beyond distributing an unchanged stock of resources. A business could develop a better product, train workers, or organize production more effectively. Understanding who benefits would then require asking how new productive possibilities arise and how people participate in them. Differences in rewards and differences in what can be accomplished belong in the same history.

A treatment entitlement raises a related problem: how does a formal claim become an appointment a person can use? Answering it requires examining the service and the practical steps through which care becomes available.

These connections give the broader question its meaning: **how do social systems change across time, and why do their trajectories diverge?** The discussion follows people and organizations whose actions affect one another through rules, resources, and relationships. Some connections support a trajectory; others interrupt or redirect it.

Empirical studies provide evidence about particular populations, interventions, and periods. Simplified examples isolate relationships, and a mathematical prototype demonstrates what can follow from stated assumptions. Cases from different settings help connect ideas rather than form a controlled comparison of entire social systems. An illustration shows how a result can arise; explaining an observed result requires evidence about the process that produced it.

Understanding that history helps locate possible changes. Evaluation asks which advantages are justified, whose opportunities improve, and which burdens are acceptable. These judgments require attention to what people can actually do, how they are treated, and what present activity leaves possible for others and for the future.

To begin, however, we must look behind the people and positions entering the first comparison. Our imagined owners already possess skills, savings, expectations, and experience. Where did those differences come from? Before following how differences become consequential, we need to understand how they were formed.

CHAPTER 1

Where Do Consequential Differences Come From?

Two young adults applying for their first full-time jobs provide a simple comparison. One has a stronger qualification, more experience, and a clearer sense of how to approach employers. These differences may matter to the hiring decision. Their application forms record positions reached through earlier experience.

Even with similar family incomes, they might have attended different schools, encountered different opportunities, or responded differently to opportunities they shared. One might have had help arranging work experience; the other might have needed to spend that time earning money.

The people entering an explanation have already lived through earlier activity. Some of what appears at the beginning of the present comparison may be the result of a previous one. Understanding individual capabilities therefore includes investigating how they developed.

Where children grow up provides a concrete way to investigate that history. Can exposure to a place change what they later become able to do? And how can we distinguish an effect of the place from differences among the families living there?

1.1 What can a map of opportunity tell us?

The Opportunity Atlas documents adult outcomes by the neighborhoods in which children grew up. Its estimates show substantial variation across childhood locations, including nearby areas and comparisons among children whose parents had similar incomes. The map makes a question visible: why do children from these places reach different adult positions? [4]

Residential selection complicates the interpretation. A family seeking a home might weigh the rent, the journey to work, proximity to relatives, and the schools available. The home it obtains also depends on what is affordable and which landlords will accept its application. Two families with the same income could consequently reach different places for reasons that also matter to their children.

Family employment connections illustrate the difficulty. They could influence both where a family settles and the opportunities available to its children. Living in the destination could also open relationships that the family lacked before. Both influences could contribute to the children's later earnings.

Separating these explanations requires evidence about what happens when similar families receive different opportunities to move, or when children experience different amounts of time in a destination.

1.2 What happens when the opportunity to move changes?

The Moving to Opportunity demonstration created such a comparison. It recruited low-income families living in distressed public or assisted housing in five U.S. cities. One randomly assigned group was

offered housing vouchers and assistance intended to support a move to a lower-poverty neighborhood; comparison groups received different assistance or no new voucher through the demonstration. The random assignment concerned the offer, not a compulsory relocation. Some families offered the opportunity moved and others did not. [5, 6]

Comparing randomly assigned offers provides a firmer basis for studying what the opportunity changed than comparing families who independently chose different neighborhoods. Estimating the effects of moves induced by the offer requires a further step; the movers themselves were not randomly selected. [5]

The findings varied by outcome and stage of life. Earlier evaluations found improvements in neighborhood safety and aspects of health, but little detectable improvement in adults' employment or earnings. Youth outcomes also differed across groups. [6–8]

Later work by Chetty, Hendren, and Katz followed children into adulthood using administrative records. Children who moved to lower-poverty neighborhoods when young—below age thirteen in the study's main comparison—experienced gains in college attendance and adult earnings. Older children did not show comparable benefits, and the estimated gains declined with age at move. The authors considered both the duration of childhood exposure and the disruption of moving in interpreting this pattern. [5]

The earlier adult results and later childhood findings concern different people, stages of life, and outcomes. Together, they show why an assessment of a move must specify what changes, for whom, and over what period.

1.3 Why does the age at move matter?

Chetty and Hendren investigated exposure beyond the MTO demonstration by studying more than seven million families moving across U.S. counties and commuting zones. Children's later outcomes moved toward those associated with the destination in proportion to how much of childhood they spent there. Their related work estimated differences in childhood exposure effects across counties. [9, 10]

These were observational studies of family moves, not randomized assignments. The researchers examined alternative explanations using comparisons among siblings, moves associated with displacement, and other tests. Their causal interpretation depends on how successfully the designs separate exposure from other differences among families and their histories. [9, 10]

A family moving with children aged eight and sixteen illustrates why exposure differs. Both children enter the same home on the same day. Yet the younger child will spend more of childhood there, while the older child brings a longer history of schooling and relationships into the move. The same event in the family's record creates different developmental experiences.

A new environment might provide opportunities that take years to use. Moving might also interrupt friendships or schooling, with different consequences at different ages. These are possible explanations of the age pattern, not separately established mechanisms. [5] The relevant history includes where a child lived, when, for how long, and what preceded and followed the move.

The findings support a causal contribution of exposure to place, with duration helping explain its consequences. [5, 9, 10]

1.4 What about the neighborhood does the work?

The MTO offer changed a residential situation rather than assigning each possible neighborhood influence separately. It therefore leaves open the individual contributions of schools, safety, relationships, stress, and other features. The mover studies likewise identify consequences of exposure to places. [5, 6, 9, 10]

A move can alter a child's school journey, classmates, access to outdoor space, and contact with nearby relatives. These influences may depend on one another. A shorter journey could make an after-school activity practical; losing help from a relative could make another activity harder.

Knowing that an environment affected a trajectory is different from knowing which parts of that environment produced the effect. A housing move, a school improvement, and support for a local organization are different actions. Evidence for one cannot automatically be treated as an estimate of the others.

Access to an activity in this example depends jointly on distance, family arrangements, and the activity's availability. What matters is whether this child can use them together. Formation already involves combinations of opportunity and support.

1.5 Can the family use the opportunity it receives?

Before a child can experience a different neighborhood, the family must be able to reach it. A voucher can help with housing costs. The family also needs to locate a suitable home, complete an application, and secure a landlord's agreement.

The Creating Moves to Opportunity experiment in Seattle and King County investigated these barriers among families receiving housing vouchers. It tested support that included information, financial assistance, and individualized help with housing search and landlord engagement. The intervention substantially increased moves to neighborhoods with higher rates of upward mobility. A further experimental comparison found that customized search assistance was especially important to the result. [11]

The measured outcome here was the move, not those children's eventual adult earnings. The earlier neighborhood studies help explain why the destination could matter. [5, 9–11]

A child's environment can therefore depend partly on the process through which the family obtains housing. Formal assistance, the search for a home, the landlord's response, the realized move, and subsequent exposure are related steps. An obstacle before the move can prevent the later experience from taking place.

To interpret a family's decision, we must consider what housing it could realistically obtain, including whether a landlord would accept its application. Its destination can reflect restricted options as well as preferences.

Policy implementation can thus influence the formation of individual differences. A child's residential experience develops through family choices and the opportunities that housing rules and organizations make usable.

1.6 How do circumstances and choices interact?

An opportunity can require effort, judgment, and persistence while being unequally available. The applicant who gained useful experience still had to do the work; access made that effort possible.

The applicants may also differ in interests, abilities, decisions, and responses to similar experiences. These remain part of the explanation alongside the contribution of neighborhood exposure established by the research. [5, 9, 10]

A skill can be useful now and also have been developed through earlier access. An expectation can influence a choice now and also reflect previous experience. Present action and the history of opportunity belong in a connected explanation.

For a child, many early circumstances are unchosen. Parents' resources, the initial home, and the timing of a move can count as luck from the child's perspective, while being connected to decisions and constraints elsewhere.

The neighborhood evidence also shows that some early circumstances can be altered, with later consequences that depend on timing. [5, 9, 10] The starting position influences a path that remains open to subsequent experience and action.

1.7 What does a later classification record?

An employer's assessment can identify which applicant is better prepared at the hiring date. Explaining how that preparation developed is another task; deciding how to reward it is another. The assessment can inform the hiring decision while leaving room to ask how the less-prepared applicant might acquire the missing experience.

This is where the chapter connects to the point-in-time puzzle. A cross-sectional comparison may take education, skills, health, or connections as current characteristics. The dynamic account asks which of those characteristics have been shaped by previous opportunities and experiences. The two accounts can describe successive parts of the same explanation.

Assessing a mobility program's effects on inequality across the population would also require examining who participates, what happens to those who do not, and how later institutions translate changed opportunities into rewards.

The people and positions entering a social process already have histories. Some differences are inherited, some developed, and some altered through action under particular conditions.

A skill may open one opportunity and be excluded from another. A resource may be usable for a season or remain under someone's control for years. A gain may be consumed, retained, or transferred. To understand which differences become consequential, we must ask what people are permitted and able to do with them.

CHAPTER 2

How Do Rules and Property Make Differences Matter?

China's Household Responsibility System expanded household control over agricultural production while farmland remained collectively owned. Beginning in the late 1970s, households gained greater discretion over farming and a stronger claim on the returns from their work. [12, 13] How could the consequences of working the land change so substantially while the ownership label remained the same?

The distinction concerns what households could decide, what they stood to gain, and how long they expected their claims to last. Land has productive possibilities; the rights attached to it determine who may use those possibilities and receive the returns. Examining those rights connects the present use of a resource with what its holder can carry into the future.

2.1 What does holding a resource allow someone to do?

A one-season cultivation right creates a different investment horizon from a claim expected to last for many years. Both allow a household to use the land. The duration of the claim matters when an improvement takes several seasons to repay its cost.

Permission to let another farmer use the field in return for payment adds an option without adding land. The ability to exclude an unauthorized user adds another form of control. Use, management, income, exclusion, and transfer can belong to different people and last for different periods. Deakin and Meng's analysis of the Household Responsibility System emphasizes this division of rights among households and collective bodies. [14]

Title is one part of economic control. Understanding a household's position requires specifying the rights it holds and what it can do with them in practice. A long-term claim, a transferable claim, and a claim on income offer different possibilities even when they concern the same field.

The distinction extends to the business discussed in the Introduction. A worker may operate a machine while its owner receives income from its use. A long-term lease may support one investment decision and a revocable permission another. What one person controls helps determine what another must obtain permission to use, pay for, or go without. Property organizes relationships among people around the use of resources.

2.2 What changed under the Household Responsibility System?

Before the reforms, agricultural production was organized substantially through collective structures associated with the people's communes. Production decisions and remuneration were collective to a significant degree, weakening the connection between an individual household's additional effort and the return that household could keep. [12, 13]

Under the Household Responsibility System, households became the principal units responsible for farming their allocated land. They gained greater discretion over production and retained more of the output or income remaining after required obligations were met. Collective ownership persisted alongside these changes in household control and claims on proceeds. [12, 13, 15]

What remains after obligations and costs are met is often called the *residual return*. A household's interest in producing another unit depends partly on how much of that return it can keep. The reform altered the connection between household decisions, additional work, and reward.

McMillan, Whalley, and Zhu examined the contributions of price changes and changed individual incentives to agricultural productivity growth after 1978. They attributed a substantial part of the increase to changes in the payment system. Lin's province-level analysis also found a large contribution from decollectivization, while distinguishing it from price adjustments and other reforms. [12, 13]

The studies estimate contributions within a historical reform, rather than a randomized experiment holding other influences fixed. Their estimates differ, but both support an important role for the changed relation between household activity and returns. [12, 13]

2.3 How can different rules change the use of the same land?

Decision authority and claims on returns affect different parts of a household's choice. Authority determines which production decisions the household can make. Its claim on the yield determines what it stands to gain. A household that holds both rights can respond differently from one permitted to choose a method but unable to retain its additional yield.

For a household working the same plot, greater discretion could make it possible to carry out an additional task at a useful time. Keeping more of the resulting harvest could make that task worthwhile. The household may use its knowledge differently or organize its work more carefully while the physical plot remains unchanged.

A farmer's skill and effort operate through these opportunities. The rules affect how skill can be used and what its use yields; the farmer still has to choose and carry out the work. Individual action and institutional conditions belong in the same explanation.

The return claim also affects exposure to risk. A household that keeps the benefit of an unusually good harvest may also bear more of the cost of a poor one. Its response will depend partly on the reserves or support available when the harvest disappoints.

Rules can change what a resource makes possible without changing the amount of that resource.

They also allocate the resulting benefits and losses. Explaining a household's return therefore requires connecting its land and effort to the decisions, claims, and support through which they become consequential.

2.4 How did a different arrangement become an accepted rule?

Deakin and Meng trace the Household Responsibility System's development through experimentation, local protection, and wider political acceptance. The 1978 agreement among households in Xiaogang became a prominent episode, but responsibility arrangements also developed elsewhere, including Anhui and Sichuan. Local practice, provincial support, and central authorization played different roles in the reform's spread. [14]

The World Bank and China's Development Research Center report that by the end of 1983 the new system covered the large majority of rural households. Particular changes in practice had become a much wider arrangement. [15]

Wider recognition alters the position of someone considering a new practice. A household acting under uncertain local tolerance faces different risks from one whose decisions have the support of broader authority. Recognition can affect expectations about continued use and the responses of other organizations.

In the historical account, local results gained influence through interpretation, protection, and endorsement. [14] Institutions acquire their effects through a history of adoption as well as through their subsequent operation. This history explains how households obtained the rights discussed here; Chapter 6 follows how reported outcomes enter later rule changes.

2.5 Does a better incentive for effort solve the investment problem?

A household might find it worthwhile to tend its current crop more carefully while being reluctant to finance an improvement that benefits several farms over many years. The first decision concerns a return the household can obtain directly. The second concerns how benefits are shared and who will hold the relevant rights when they arrive.

Choe develops this distinction in examining China's rural reform. His analysis argues that household residual claims strengthened work incentives but could leave inadequate incentives for investments whose benefits extended beyond the investing household. In Choe's account, the productive gain coexisted with a further problem of investment and coordination. [16]

A shared drainage channel illustrates the difficulty. One household could pay for an improvement from which neighboring fields would also benefit. Even when the total benefit exceeds the cost, the paying household might recover too little to make paying alone worthwhile. A stronger claim on its own harvest leaves the financing of the shared channel to be resolved.

Investment also depends on the duration and security of control. A household expecting to lose a plot before an improvement pays for itself has less reason to finance that improvement. Someone seeking to revise the allocation of land may value flexibility instead. The rules governing duration and reallocation must address both interests.

Small, separate plots raise a related issue of scale. Individually organized production may still benefit from joint investment, shared services, or combining land use. These are possible ways to coordinate activity across households; the conditions for making each work require examination.

The HRS discussion therefore concerns several connections: household effort to household returns, investment costs to shared benefits, and present control to future claims. Strengthening one connection leaves the others to be organized. Private and collective decisions can have different roles within the same productive activity.

2.6 Which consequences can be carried forward?

China's rural property arrangements continued to evolve after the initial reform. The 2022 World Bank report describes extensions of contract duration and later changes separating ownership, contracting, and

management rights and enabling land-use transfers. Collective ownership persisted as these other rights developed. [15]

Duration, transfer, and retention each carry consequences forward in a different way. A durable claim provides a longer horizon for planning. A transferable right permits exchange. Retained proceeds remain available for later use. These possibilities depend on the rights attached to the particular resource.

Equipment bought with a household's retained earnings makes the connection visible. The harvest has ended, but part of its result is available for the next season. Where the household also has the right to transfer that equipment to a child, the gain reaches someone who did not produce it. The equipment example concerns rights of retention and transfer; households' rights over collectively owned land remain those specified by the agricultural arrangements.

This is why property matters to the inequality question. Two people with similar current earnings can hold different stocks of assets or claims on future income. Those positions may reflect earlier activity or transfers. Explaining them requires following what could be retained and passed on, alongside the skill and effort supplied in the present.

Assets can also deteriorate or lose value, and claims can be limited, contested, or revised. The history includes both the routes that preserve an advantage and the events or decisions that interrupt it.

A right can shape the return to an action today and the resources available for another action tomorrow. Chapter 6 follows this connection through repeated activity.

2.7 From rights to realized benefits

The land remained collectively owned while household decision authority and claims on returns changed. Those changes helped alter productive behavior. [12, 13] Longer-run investment, coordination, and security remained consequential alongside the initial gain. [15, 16]

China's subsequent transformation also involved price reform, modern inputs, infrastructure, industrial development, and opportunities outside agriculture. Higher agricultural productivity could make resources available for other activity, while productive nonfarm work required jobs and supporting arrangements. [15] Chapter 4 follows that transition.

For our broader inquiry, rights connect the characteristics of people and resources to the benefits they receive. Equally skilled farmers on identical plots could receive different returns under different rules. A sharing arrangement could instead moderate differences in returns when production differs. The distribution received depends on how skills and resources operate through decision rights and claims on proceeds.

Higher production supplies one consideration for evaluation. Control, security, and the burdens of the arrangement supply others. Chapter 7 considers how these judgments bear on what counts as improvement.

A claim must also be recognized and exercised. A household may hold a use right while facing a dispute over access; a person may qualify for a public program while struggling to obtain its benefit. Chapter 3 follows the decisions and organizations through which a formal entitlement becomes something a person can actually use.

CHAPTER 3

Who Gets What—and How?

Two people who qualify for the same health-insurance program can follow different paths. One enrolls, remains covered, and obtains the appointments needed. The other encounters a problem completing renewal or cannot find a listed provider accepting new patients. The benefits each can use depend on what happens between eligibility and care.

Medicaid makes that path visible through enrollment, renewal, provider networks, and appointments. Insurance also provides financial protection, a benefit that can arise without a measurable improvement in health. [17, 18] Following these connections means examining **who receives what, which actions make it available, and who bears the remaining work and risk.**

3.1 What has been allocated when someone qualifies?

An eligibility rule specifies who qualifies. Providing the benefit requires further action: the agency must establish eligibility, the person must become enrolled, and coverage must remain in effect. When treatment is needed, the person must find a suitable provider and obtain an appointment. A university's selection of applicants visibly allocates places; a public insurance program allocates benefits through a sequence of administrative and service decisions.

That sequence also distributes work. Requiring applicants to locate documents and respond to notices places tasks on them. Verifying information within an agency places more of the work on the administration. Insurance, meanwhile, redistributes the financial consequences of illness. The Oregon evidence discussed below found that coverage reduced households' exposure to medical financial liabilities as well as increasing service use. [17, 18]

Coverage can protect against a possible bill before treatment is needed. Later use of services may affect both finances and health, with different results and timing. Realized allocation therefore includes continuing protection, services received, and the burdens of obtaining them.

3.2 What does obtaining insurance change?

People who enroll in Medicaid can differ from those who do not in health needs, knowledge, and ability to navigate the program. A direct comparison of their later outcomes would mix the effects of insurance with those pre-existing differences. The Oregon Health Insurance Experiment supplied a stronger comparison.

In 2008, Oregon used a lottery to select people from a waiting list for the opportunity to apply to a limited Medicaid program for low-income uninsured adults. Selection increased the probability of insurance coverage by about 25 percentage points. Researchers used this randomized variation to estimate the effects of coverage induced by the lottery. The assignment concerned the opportunity to apply, not automatic enrollment; the findings concern adults who had sought access to this particular program. [17]

During the first year, coverage increased medical use and reduced financial strain, including out-of-pocket spending and the risk of unpaid medical bills being sent to collection. Approximately two years after the lottery, a follow-up examination found greater use of services, including preventive care, and increased diabetes diagnosis and use of diabetes medication. It also found fewer positive screenings for depression and substantial protection against catastrophic out-of-pocket expenditure. [17, 18]

The follow-up did not find statistically significant improvements in blood pressure, cholesterol, or glycated hemoglobin, a measure used to assess blood-glucose control. This finding concerns the selected measures over the observation period; it does not establish that every health effect was zero. [18]

The combined findings distinguish financial protection, service use, and health. Reduced financial strain is a benefit in its own right. Fewer positive depression screens identify a mental-health gain. Increased service use records greater utilization, while its health consequences require their own assessment. These outcomes cannot be replaced by a single verdict about whether insurance worked.

Timing also affects what can be observed. Financial exposure may change promptly, while some health effects may require different treatment or longer observation. Establishing any later gain would require evidence from that later period.

3.3 Who must act to keep coverage in place?

Renewal brings an existing benefit back into an administrative decision. During the COVID-19 emergency, a continuous-enrollment provision largely prevented Medicaid disenrollment. States resumed regular disenrollments in 2023 as they redetermined eligibility. This return to ordinary renewal processes, often called the *unwinding*, generated records of who remained covered and why coverage ended. [19, 20]

KFF's archived summary, updated September 12, 2024, recorded more than 25 million disenrollments and more than 56 million renewals. Among disenrollments with reported reasons, 69 percent were classified as procedural rather than as determinations of ineligibility. These historical renewal records cover different reporting periods across states, with varying detail; they are not current enrollment totals. [20]

A procedural termination records failure to complete the required renewal process. It leaves the person's substantive eligibility unresolved in that record. The distinction matters because an incomplete response and a finding of ineligibility identify different reasons for ending coverage. [19, 20]

A notice-and-deadline process makes continuation depend partly on the beneficiary's response. Coverage could end after an incomplete response even when the underlying income has not changed. If the agency can verify the necessary information from records it already holds, the beneficiary may remain covered without taking that additional step.

Verification using available information without a separate beneficiary response is called *ex parte* renewal. A 2025 study examined federal assistance to improve such renewals in four states. Relative to other states, *ex parte* renewals increased by 21.6 percentage points, overall renewals by 7.7 points, and procedural denials declined by 8.3 points. The study compared state administrative data; the four states were not randomly assigned. [21]

The findings connect administrative capacity with coverage continuity. [21] Verification remains necessary, but its organization determines what information the agency uses and how much work falls on the person whose coverage is at stake. An administrative change can preserve coverage by carrying more of the verification task within the agency.

3.4 Can a covered person obtain an appointment?

Coverage determines the financial terms of seeking care under a plan. Obtaining that care also requires an available provider. Studies of networks and appointment requests examine different parts of that practical access.

Wallace, Lollo, and Ndumele's 2020 study, using 2017 data from five states, found that the average Medicaid managed-care network included more office-based physicians than the average exchange-plan network in the same geographic markets. Participation estimates were higher when based on directories than when based on physicians' survey answers. These findings concern network inclusion; an appointment request tests whether an included provider can actually see a new patient. [22]

Appointment requests were the focus of an investigation published in 2014. The U.S. Department of Health and Human Services Office of Inspector General (OIG) drew a stratified random sample of 1,800 primary-care and specialist providers listed by Medicaid managed-care plans. After exclusions and nonresponse, its analysis used 1,092 provider observations from calls made in 2013. [23]

Projecting the results to the represented provider population, OIG estimated that slightly more than half could not offer an appointment: 35 percent could not be found at the listed location, 8 percent said they did not participate in the plan, and another 8 percent were not accepting new patients. Among those offering appointments, the median wait was about two weeks; more than one-quarter had waits exceeding a month. [23]

An inaccurate location, nonparticipation in a plan, and a closed patient list each interrupt an appointment request in a different way. They call for different responses. The percentages are provider-level estimates for the study's sampling frame and period, not beneficiary-level measures or a current national access rate.

A later OIG evaluation, published in 2026, examined maternal-health directories for plans associated with three large managed-care parent companies in five states. It found listings for providers who were not in network, missing listings for some in-network providers, and inaccurate contact details. The evaluation concerns the information available to someone seeking care, rather than subsequent treatment or maternal-health outcomes. [24]

Together, the studies distinguish the breadth of a network, the accuracy of its information, and the availability and timing of appointments. Payment can also influence providers' decisions. Using simulated-patient calls across ten states, Candon and colleagues found that changes in Medicaid primary-care fees were associated with changes in appointment availability for new Medicaid patients. This observational comparison concerns the period examined. [25]

3.5 Where does an organization change the outcome?

The beneficiary encounters the combined operation of several organizations. In the arrangements examined here, agencies determine and renew eligibility, managed-care plans organize networks, and provider practices decide whether and when appointments are available. [19, 21–25]

Each organization acts on a different part of the path. Automatic renewal can preserve coverage while an unavailable appointment prevents care. A broad provider network can offer options while an incomplete renewal interrupts insurance. Locating the obstacle helps identify a response: better information systems could support verification, a corrected directory could prevent wasted calls, and additional clinical staff could expand appointment capacity.

The work required of beneficiaries also matters. Two equally eligible people receiving the same notice may have different time, internet access, or help resolving an error. The same instruction can therefore demand different effort and lead to different coverage outcomes. Personal resources become consequential through the tasks the process requires; revising those tasks can alter the consequences of unequal support.

The renewal study provides evidence of an administrative change associated with more continuing coverage. The provider audits locate obstacles between a listing and usable care. [21, 23, 24] The relevant explanation follows each organization's actions and what those actions make possible for the beneficiary.

3.6 What does “who gets Medicaid” leave out?

An enrollment count brings together people with different experiences. A person may remain insured while waiting for care, obtain treatment and avoid a large bill, or lose coverage during renewal without a determination of substantive ineligibility. Following what each receives requires measures of continuity, protection, access, and health.

The studies make those distinctions observable. Oregon's randomized insurance opportunity supports estimates of the effects of induced coverage on service use, financial protection, and measured health. [17, 18] Administrative records distinguish renewals from recorded reasons for termination, while comparisons of renewal procedures investigate continuity. [19–21] Directory audits test listed information, and appointment studies examine whether providers offer care. [22–25]

Their evidentiary reach differs. An appointment audit leaves open whether a person later obtains care elsewhere. A procedural-termination record leaves substantive eligibility unresolved. The studies also concern different populations, states, and years, rather than one common set of beneficiaries followed through every stage. Their percentages cannot be combined into a single national probability of reaching care.

The connection to unequal outcomes is now concrete: **people meeting the same eligibility rule can encounter different administrative demands, provider options, and practical support.** The distribution of benefits reflects how those circumstances operate together. Classification identifies who qualifies; the work of agencies, plans, and practices helps determine what follows.

3.7 What remains a question of judgment?

Applying the same substantive eligibility rule and giving eligible people a reasonably comparable ability to exercise their entitlement are distinct concerns. Automatic verification and a beneficiary-response requirement can distribute work differently while serving the same eligibility decision. Judging that division of work requires attention to administrative cost, reliable verification, privacy, and responsibility for supplying information.

Oregon's evidence of financial protection supplies a reason to value coverage. [17, 18] Assessing its priority also requires considering other uses of resources. Identifying an access barrier gives a problem to address; a proposed remedy must be assessed for its ability to remove that barrier and for the costs and burdens it introduces.

The descriptive and evaluative tasks remain connected. Establish what people receive, what they must do, and where the process affects their opportunities. Then ask which differences and burdens are defensible and which should be changed. Chapter 7 develops these reasons for judgment across the cases.

3.8 From realized access to the ability to provide

Verification and renewal support continuing coverage. Provider organizations affect access to care when it is needed. Time, information, support, and provider options affect what a person can obtain at different steps. Following those steps connects individual circumstances with the decisions of agencies, plans, and practices.

Appointment availability also brings productive capacity into view. Correct information can help someone reach care already available. Providing additional care may require more clinicians, facilities, or better organization. Helping a person reach an existing service and developing the ability to supply more service are distinct tasks.

That ability evolves. People acquire skills, organizations learn or lose the capacity to coordinate, and equipment and technology expand or restrict what can be accomplished. The benefits available for allocation change with them. Chapter 4 follows how the capacity to produce and provide those benefits is built, maintained, and reorganized.

CHAPTER 4

How Do Social Systems Build— and Reallocate—Capacity?

Better scheduling could help a health service redistribute appointments and reduce unused clinical time. Training additional staff could improve future provision while taking time away from today's service. The same action can affect both who obtains care and how much care the service can provide.

This chapter follows changes in the **amount, quality, and organization of what people can produce, provide, and sustain**. Industrial robots make the connection concrete: a machine can expand productive possibilities while changing the tasks, organizations, and opportunities through which people participate.

4.1 Capacity is an ability, not simply a quantity of resources

Capacity means the ability of people and their organizations to carry out activities and sustain results under particular conditions. The relevant ability might be producing food, delivering treatment, maintaining infrastructure, developing knowledge, or coordinating a response to disruption.

Equipment becomes usable through staff, knowledge, and working arrangements. An organization with sophisticated equipment but too few people able to operate it faces a different constraint from one that needs a better way to coordinate its existing resources. Understanding capacity requires examining how those resources work together.

Observed output records what was accomplished during an interval. Capacity concerns what can be accomplished and sustained. A service might temporarily produce more by exhausting its staff or postponing maintenance. Another might produce less during an investment period that improves what it can subsequently deliver.

People bring abilities formed through earlier experience; authority, resources, and working arrangements shape how they can use them. An organization combines these elements, while allocations of training, finance, and equipment affect what it can do next.

4.2 A new technology changes possibilities before it changes outcomes

For a manufacturer considering a robot, performing a task is one part of adoption. The machine must also fit into a functioning production process. Workers may lose a task, take on a revised role, or find their skills more useful in the new combination. Following adoption means examining the machine's performance, the organization of production, and the resulting opportunities for people.

Acemoglu and Restrepo's task-based analysis distinguishes two processes. Automation transfers some tasks previously performed by workers to capital, potentially raising productivity while reducing demand for labor in those tasks. Creating new tasks in which people have a comparative advantage can increase demand for their work. The authors call these processes *displacement* and *reinstatement*. Their balance

depends on the tasks actually displaced and created; sufficient new tasks do not follow automatically from automation. [26]

Graetz and Michaels examined industries in seventeen countries between 1993 and 2007. Their estimates indicate that increased robot use contributed to labor-productivity growth, increased total factor productivity, and lowered output prices. They did not find a statistically significant reduction in total employment in their analysis, but they did find a reduction in the employment share of low-skilled workers. [27]

Labor productivity measures output per hour worked. Total factor productivity relates output to combined inputs, including labor and capital. A gain in the second measure means producing more relative to the measured inputs taken together. [28]

Productive gains and an uneven distribution of employment effects can therefore occur together. The consequences depend on adoption, the reorganization of work, competitors' responses, and the opportunities available to affected workers.

4.3 An expanding firm is not the same as an expanding opportunity for everyone

Acemoglu, Lelarge, and Restrepo's study of French manufacturing firms shows why the location of a gain matters. Robot adopters increased productivity, value added, and overall employment, while the labor share and the share of production workers in their employment declined. The authors also find that adopters expanded at competitors' expense, producing an overall negative association between robot adoption and employment in their wider analysis. [29]

A manufacturer that becomes more productive might lower prices, attract customers, and expand production. A sufficiently large expansion could increase its employment even while fewer workers are needed for each unit of output. Where those customers previously bought from competitors, the competitors could contract, reduce hiring, or close. The adopting firm's employment record captures one part of this wider adjustment.

The location of a gain and the extent of a gain are different questions. Technology can expand what a set of resources produces and shift where production occurs or which organizations control it. Both changes matter to the people participating.

Particular organizations adopt, delay, experiment, or decline. Their choices affect other organizations, including those that retain existing technology. Workers, suppliers, and communities can consequently experience changes initiated elsewhere. Organizations turn a technical possibility into operating capacity, with effects that extend through these relationships.

4.4 Existing workers and people seeking entry may face different changes

Acemoglu and Restrepo's study of U.S. local labor markets found negative effects of industrial-robot exposure on employment and wages. Its local labor-market comparisons capture consequences beyond employment inside adopting firms. [30]

The German evidence examined by Dauth and colleagues shows a different adjustment pattern. They find manufacturing displacement associated with robot exposure, offset by additional service employment.

Established workers experienced greater employment stability within firms, often by taking on new tasks in their original plants. Much of the adjustment burden fell instead on young people entering the labor market. [31]

Fewer opportunities can appear as reduced hiring of entrants, while existing workers remain employed in altered roles. The German study makes these routes visible. [31] Following employment therefore requires asking who remains inside an adapting organization, who changes employer or occupation, and who finds that a previous route into work has narrowed.

The studies examine different settings, units, and adjustment processes. Their contrasting findings do not isolate which national institutions account for the differences. [27, 29–31] They direct attention to the resources and institutional arrangements supporting each transition.

Capacity and allocation meet in these changes. Reorganizing production also reorganizes the routes through which people participate in it and receive its returns.

4.5 Where can the released labor go?

China's Household Responsibility System begins with a change in rights and incentives. As Chapter 2 discussed, McMillan, Whalley, and Zhu attributed a substantial part of the post-1978 agricultural productivity increase they examined to changes in the payment system associated with the responsibility system. [12] That productive gain opens a further inquiry into work beyond farming.

A household able to produce a given crop with fewer days of work could expand farm production, reduce working hours, or use some of the time elsewhere. Earning more outside farming requires another productive activity that the household can reach.

The World Bank and the Development Research Center of China's State Council describe the expansion of township and village enterprises as an important part of this transition. From 1978 to the mid-1990s, these enterprises provided nonfarm jobs for large numbers of rural workers. Employment close to home also allowed households to combine farming with wage work rather than requiring an immediate, complete departure from agriculture. [15, p. 18]

The same account connects enterprise expansion to demand for consumer goods and to local governments' incentives to develop production. Available labor was one part of a transition that also depended on organizations and markets capable of using it. [15, p. 18]

A village workshop illustrates the connection. A household may have time for additional work while the workshop lacks finance to purchase materials. Even with materials, the workshop needs buyers. Agricultural productivity creates a possibility whose realization depends on capacities outside farming.

An improvement in one activity can create a possibility that other capacities are needed to realize.

The same issue arises when automation reduces the labor required for a task. What affected workers can do next depends on the productive opportunities and access available elsewhere.

4.6 Capacity building and adjustment run on different clocks

Installing equipment, learning to use it, and finding new roles for workers take time. In an automation project, some workers might move into revised roles immediately while others need training. Someone seeking a first job may encounter a changed hiring pattern while the firm and local labor market are still adjusting.

These processes run on different schedules and involve different people. A firm may resume expansion while displaced workers are still seeking employment. Later jobs may go to new entrants rather than those who lost earlier opportunities. Following adjustment requires tracking these transitions as well as the firm's eventual output.

Resources available during the transition also affect its course. Two workers needing a year of training could face very different choices: one has savings and access to a suitable course; the other must take immediately available work and cannot attend. Their paths depend on the means of crossing the transition as well as the opportunities at its end.

4.7 Why does the outcome distribution not mirror its components?

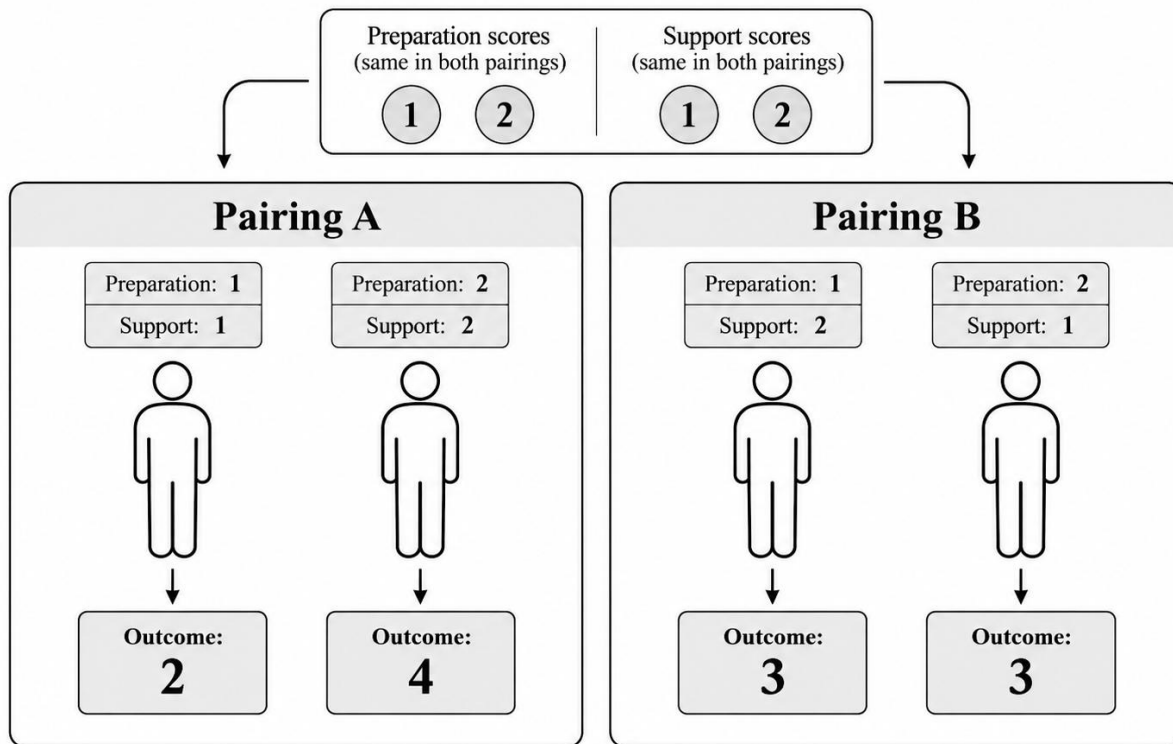
What a productive arrangement can accomplish and how its proceeds are divided jointly shape the distribution we observe. Examine the two business owners' earnings during the same year, before either business closes. The point-in-time view asks how their skills, resources, relationships, and working arrangements produce what each receives during that period.

Alongside the distribution of skills, we need to know who has the equipment and access to use them. The more skilled owner might also have better equipment, or equipment might compensate for a difference in skill. The separate distributions leave these combinations unspecified.

To isolate the role of pairing, give two people preparation scores of 1 and 2, and assign the available support amounts scores of 1 and 2. Define an outcome score as preparation plus support. Pairing the lower values together and the higher values together gives outcomes of 2 and 4. Pairing lower preparation with higher support, and higher preparation with lower support, gives 3 and 3.

Both pairings have the same separate component distributions, the same outcome rule, and the same total outcome. **Only the pairing changes, yet the outcome distribution changes with it.**

Figure 1. Same components, different pairings



Preparation + support = outcome in both panels. The totals are unchanged; only the pairings differ.

The *joint distribution* describes how the characteristics occur together. In the business example, it concerns which owner has which combination of skills, equipment, finance, and customer access. Usable resources can include access through others: a reliable shared facility could supply equipment to someone unable to purchase it.

The way components work together also matters. A skilled worker and a functioning machine may both be needed to complete an order. Bringing them together creates a productive possibility that neither supplies alone. Organization makes the combination usable.

Rewards depend on a further set of decisions and claims. Two businesses might submit proposals that perform almost equally well in a customer's trial. Awarding the entire order to one supplier turns a small performance difference into a sharp revenue difference. Dividing the order between them would distribute the same revenue differently.

An accurate selection can therefore produce sharply unequal rewards. The size of the order matters too. Additional equipment and employees could let the selected firm fulfill a much larger order, increasing revenue without a corresponding increase in the owner's personal skill. The scale of the productive activity helps determine the return.

The division of proceeds then determines who receives it. In our business example, workers might receive agreed wages while the owner receives what remains after wages and other costs are paid. Holding those payments fixed for the current contract, additional revenue goes to the owner. Sharing the surplus would

distribute the same proceeds differently. The owner's income depends partly on a claim over a joint activity.

The other business's opportunities depend on these decisions as well: an order awarded to one supplier is unavailable to the other. In the French robot study, adopters' expansion at competitors' expense made this connection visible. [29] Demand can also expand. A customer might buy more when lower costs make a product more affordable, or a manufacturer might supply a product previously unavailable. Competition, cooperation, and demand connect participants' outcomes in different ways.

Some connections moderate differences. Shared equipment can reduce the consequences of unequal ownership. Accessible training can remove an obstacle to participation. A different division of proceeds can spread the gains from joint activity. Each operates at a particular point between a person's starting position and the benefit received.

An outcome distribution reflects which capacities and resources occur together, what those combinations make possible, and how access, selection, scale, and claims turn activity into benefits and burdens. A modest capability difference can therefore coexist with a large reward difference, while other arrangements can narrow the gap. Explaining the shape of an observed distribution requires evidence about these connections and how they developed.

4.8 What history lies inside the snapshot?

The business's equipment and customer record help explain its current earnings. Equipment might have been bought with earlier proceeds, shared through an association, or acquired before the owner entered the business. Similar present resources can come through different histories.

Current equipment helps explain what the business can do now; its acquisition history explains how that capacity became available. Following acquisition, use, and retained proceeds connects the present position with earlier activity and the possibilities for what follows.

Allocation affects the conditions under which capacity develops; changes in capacity alter what can subsequently be allocated and who is positioned to obtain it. A customer's withdrawal, equipment failure, or interruption in funding can then put that capacity under strain.

Similar capabilities can encounter different events, and the same event can meet different reserves, obligations, and options. Chapter 5 follows how these encounters redirect an apparently similar starting position toward a different path.

CHAPTER 5

Why Do Similar Starting Points Diverge?

A business can retain working equipment, skilled employees, and a sound production method yet cease operating when it cannot finance the next order. The capacity described in the previous chapter depends on funding and other support that keep it usable.

Two manufacturers entering a downturn with similar equipment and recent sales may have different reserves, lenders, customers, and payment deadlines. One might continue operating while the other closes. Their responses and the decisions of others can create further differences during the downturn.

The financial crisis of 2007–2009 gives this problem a wider setting. Housing and mortgage losses developed into financial disruption far beyond the original loans. [32] Following that expansion means examining how a disturbance reaches particular firms, what responses remain available, and how their timing redirects the path that follows.

5.1 A loss does not tell us who can absorb it

Two organizations each hold assets worth 100. One owes 60, leaving its owners with equity of 40. The other owes 95, leaving equity of 5. Hold those debts fixed while both portfolios fall in value to 96. Each organization loses 4, but the first organization's equity falls from 40 to 36, while the second's falls from 5 to 1. The same asset loss removes one-tenth of the first equity cushion and four-fifths of the second.

This is the arithmetic of leverage: financing assets with debt makes the owners' remaining stake more sensitive to changes in asset value. The organizations began with equal assets and different abilities to absorb losses. Their ability to continue also depends on payment deadlines, continued lending, and the proceeds available from asset sales.

An organization expecting payments over several years may still have a large borrowing due tomorrow. Meeting that obligation requires cash from reserves, replacement funding, or an asset sale. An otherwise comparable organization whose borrowing remains in place for several years has more time to respond.

A resource position therefore includes claims against the resources, contractual deadlines, and access to other organizations. Similar productive capacity can be sustained or lost through different obligations and sources of support. **The disturbance matters, but so does the position in which it is encountered.**

5.2 Connections can spread losses—and change their size

Mortgages could be pooled and their payment streams sold as securities, a process called *securitization*. Financial institutions could then use securities as collateral for borrowing. These arrangements connected the original loans to investors and funding markets. [33]

One important form of funding was the repurchase agreement, or *repo*: economically, a collateral-backed loan, often for a short period. Gorton and Metrick document rising collateral discounts, known as

haircuts, in the repo transactions they study. Larger discounts meant that borrowers could obtain less funding against the same valued collateral. Their evidence connects concerns about financial trading partners' ability to meet their obligations and about collateral markets to pressures extending beyond subprime mortgage assets. [33]

In *tri-party repo*, a third institution manages the collateral and carries out the exchange of cash and securities for the borrower and lender. [34]

Copeland, Martin, and Walker report comparatively stable collateral margins and funding for most borrowers in tri-party repo, while Lehman's funding in that segment fell sharply in September 2008. They challenge the claim of a system-wide repo run. [34] This is a scholarly disagreement about the reach of the repo-run interpretation, informed by evidence from different parts of the market. Rising haircuts and outright withdrawal of funding are distinct routes to funding pressure; establishing their prevalence matters to how broadly that interpretation applies. [33, 34]

Funding pressure can also enlarge losses elsewhere. An institution needing cash may sell assets. When many constrained holders sell into a market with limited buyers, prices can fall further. Lower prices weaken other holders' balance sheets and borrowing capacity, prompting additional sales. Brunnermeier explains this interaction between funding difficulties and asset sales as a mechanism of crisis amplification. [32]

Selling may relieve one institution's immediate shortage while worsening others' access to funds. The response to an initial loss has become part of the disturbance faced elsewhere.

Connections can also help absorb losses. In the theoretical model developed by Acemoglu, Ozdaglar, and Tahbaz-Salehi, greater financial interconnectedness helps absorb sufficiently small shocks but transmits distress when shocks exceed a threshold. [35] The model makes the consequences of interconnectedness depend on the size of the shock. Following the crisis requires examining both the original losses and the responses through which exposure spread.

5.3 Why firms and workers outside finance were affected

A manufacturer may need credit to buy materials and pay workers before customers pay for completed orders. Its production method can remain sound while its financing becomes unavailable. The lending relationship is part of what keeps the factory operating.

Chodorow-Reich examines this connection using banking relationships and employment data for approximately 2,000 nonfinancial firms. Firms connected before the crisis to less healthy lenders were subsequently less likely to obtain loans, paid higher interest rates when they borrowed, and reduced employment more. The employment effect was evident for small and medium-sized firms in the sample, but was not established for the largest or most transparent firms. [36]

The study uses differences in lender health to investigate whether credit availability affected employment. It also documents costs to borrowers of changing lenders, which could make an existing relationship difficult to replace promptly. [36] A relationship that ordinarily supported production could thus transmit financial distress to a firm with functioning machines and skilled workers.

Customers' reduced spending provided another route. Mian, Rao, and Sufi find larger spending responses to housing-wealth losses in U.S. areas with poorer and more leveraged households. The distribution of debt and wealth losses helps explain the uneven decline in consumption. [37]

Mian and Sufi connect household housing-net-worth losses to employment. Counties with larger losses experienced larger employment declines in businesses serving local demand. Their analysis examines competing explanations, including construction exposure and credit-supply tightening. Within the period studied, it finds little evidence that the hardest-hit counties quickly offset the shock through outward migration, wage adjustment, or expansion of employment producing tradable goods. [38]

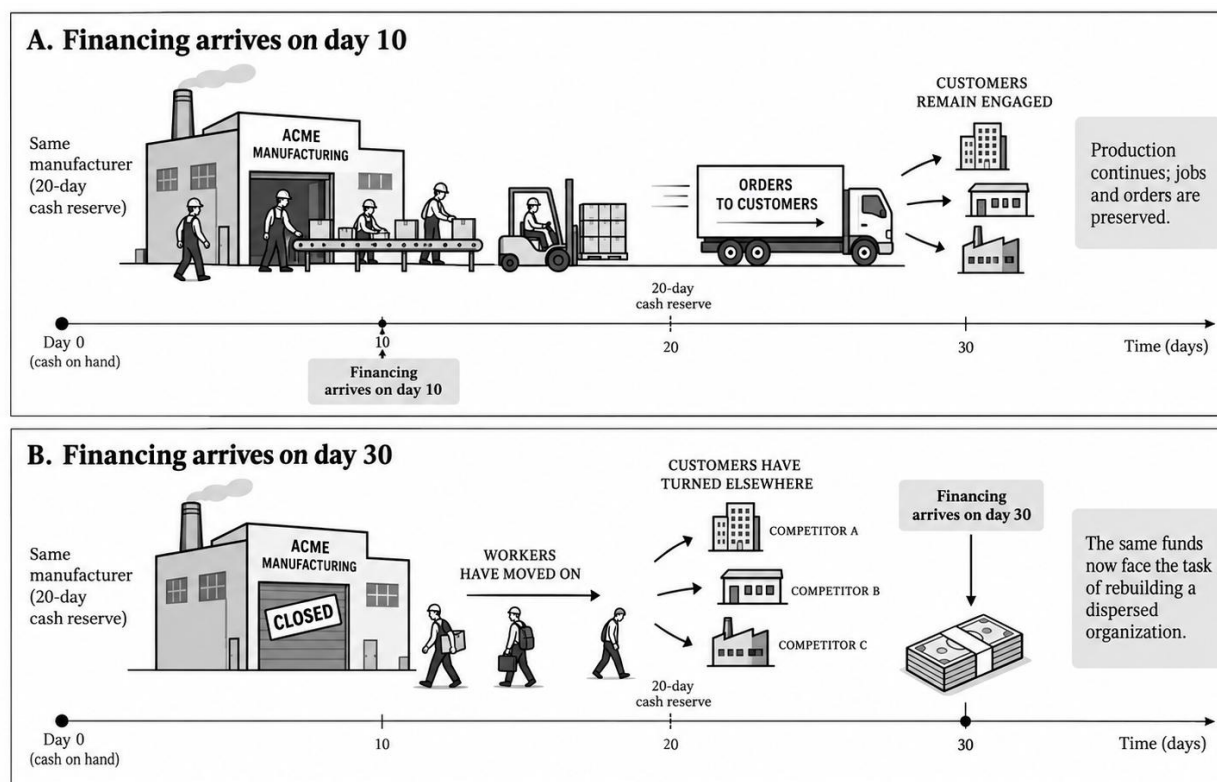
The manufacturer could face a funding problem, a loss of customers, or both. The studies investigate distinct links between financial disruption and activity outside finance; their estimates cannot simply be added into a complete account of the crisis. [36–38] For a particular firm, we need to establish which relationships transmitted the pressure and what alternatives remained.

5.4 The order of events changes what remains possible

The manufacturer's cash reserve sets a deadline for response. With enough cash to maintain production for twenty days, financing received on day ten might preserve jobs and allow existing orders to be completed. The same financing on day thirty might arrive after closure, when workers have found other jobs and customers have made alternative arrangements.

The earlier financing supports continued operation; the later financing faces the task of rebuilding an organization whose relationships have dispersed. The amount supplied can be identical while the task changes. A ten-day financing decision also has different consequences for a business with twenty days of reserves and one with only two.

Figure 2. The same support, at different moments



In this hypothetical comparison, the same financing arrives before or after a twenty-day cash reserve is exhausted. Keeping an organization operating and rebuilding it after closure are different tasks.

Public responses to the financial crisis addressed several pressures. The Federal Reserve introduced facilities to provide short-term liquidity to financial institutions and support key credit markets. These interventions targeted funding and market-functioning problems. [39]

In May 2009, U.S. banking agencies published the results of the Supervisory Capital Assessment Program for the nineteen largest bank holding companies. The assessment estimated losses, available resources, and additional capital buffers under an adverse economic scenario. Its stated purpose included sustaining lending even if the downturn became worse than expected. [40]

Access to funds, resources for absorbing losses, and information about possible future losses address different aspects of a bank's position. The interventions sought to alter those aspects through different measures. Public measures and private adjustments occurred together, however, so improvement after an announcement alone cannot establish a program's effect. Attribution requires evidence about what the response changed, for whom, and while which options remained available.

5.5 One crisis does not have one ending

An interruption can be abrupt even when vulnerability has accumulated gradually. The manufacturer's reserves may shrink while the factory continues operating. A payment it cannot meet makes the difficulty visible after the resources for responding have already been depleted.

Recovery likewise has several turning points. The U.S. recession ended in June 2009 according to the dating reported by the Bureau of Labor Statistics, but unemployment continued rising and reached 10.0 percent in October. Payroll employment reached its low point in February 2010. [41] The indicators record different parts of the economic experience.

In May 2010, Bernanke reported improved financial positions and market access among the banks assessed in the 2009 stress exercise. In the same speech, he acknowledged that bank lending continued to contract and that the intended return to a better lending environment had not yet been achieved. Stabilizing institutions and restoring credit flows were separate accomplishments. [42]

Recovery therefore needs a specified measure and benchmark. Resumed funding, renewed hiring, and restored household savings concern different parts of the history. A business that regains its pre-downturn sales after three years reaches one benchmark. Whether it also recovers the growth it would otherwise have achieved, and whether the same workers share in renewed activity, remain further questions.

5.6 Timing matters within a life, not just within a crisis

Timing also concerns the stage of a person's life at which an opportunity arrives and the experience that follows it. In Chapter 1's Moving to Opportunity study, gains in later college attendance and earnings were larger for children who moved when young, with estimated benefits declining as age at move increased. The researchers consider both exposure to a better environment and the disruption of moving. [5] The move's consequences develop through the period of life that follows it.

The training problem in Chapter 4 makes a related distinction. Training completed while someone remains in an organization can support a different transition from training offered after the person has left. Comparable resources supplied at different moments can meet different opportunities to use them.

A worker entering an occupation just before its opportunities contract may reasonably describe the timing as bad luck. The contraction can arise from decisions made elsewhere while remaining beyond the worker's control. Luck here identifies an unchosen encounter with an event whose causes can still be investigated. Explaining its consequence requires tracing the event, its timing, and the resources and options available to the person affected.

5.7 What carries an earlier event into a later period?

Closure can leave the manufacturer with a dispersed team and broken customer relationships. New financing may make production possible again, but reassembling those relationships takes further work. The earlier interruption continues to matter through what it altered.

The literature on *path dependence* examines how earlier events and choices change later possibilities and the costs or returns of taking different paths. Pierson develops one influential account through increasing returns: taking a path can create investments, learning, and coordination that make continuing along it more attractive and reversing it more difficult. Timing and sequence help explain which path develops. [43]

Persistent weakness can have another explanation: customers may still be spending less. Distinguishing continuing adverse conditions from a consequence carried forward by a completed shock requires

evidence about what changed, what persisted, and how it affected later activity. Persistence alone leaves these explanations unresolved.

An interruption can also leave new knowledge and practices. In his 2010 account, Bernanke described incorporating lessons from the stress assessment into ongoing supervision, including more cross-firm comparisons and forward-looking analysis. The changes he described concerned supervisory methods, information, and working arrangements. [42] The response could thus alter how later decisions are made; the effectiveness of the new practices remains a separate question from their adoption.

5.8 Divergence is a process to explain

The manufacturers' equipment and initial sales describe only part of their ability to continue. A lender's condition, a customer's lost income, or the arrival of support can make the same productive resources usable in one firm and unusable in another. The path develops as these events meet particular obligations and options.

Some differences were present before the downturn; others arose through responses and decisions during it. The comparison of endpoints records both together. Following the sequence helps establish which differences mattered, when they became consequential, and what could still be changed.

After the immediate episode, depleted reserves, dispersed working relationships, and revised organizational practices can remain. **An outcome can change the resources, relationships, and responses available for subsequent activity.** Chapter 6 follows how those consequences enter the conditions of the next round.

CHAPTER 6

How Do Outcomes Become Later Conditions?

An imagined farming community prepares for another growing season. Some households have money for repairs; others are short of funds. A maintenance dispute remains unresolved. Farmers at the end of the canal remember whether water reached them when it was promised. The harvest is over, but its material and organizational consequences are still present.

Nepal's irrigation systems give this connection a concrete setting. Ostrom and Gardner examine the relationship between farmers near a canal's water source and those farther downstream, including how their different positions affect bargaining over maintenance and water use. [44]

A successful harvest, an agreed repair, and a routine allocation can each affect what people are able and willing to do next. Following one season into another lets us examine **how operating an arrangement changes the resources, expectations, and relationships through which people maintain or revise it.**

6.1 The next round does not begin from the same starting point

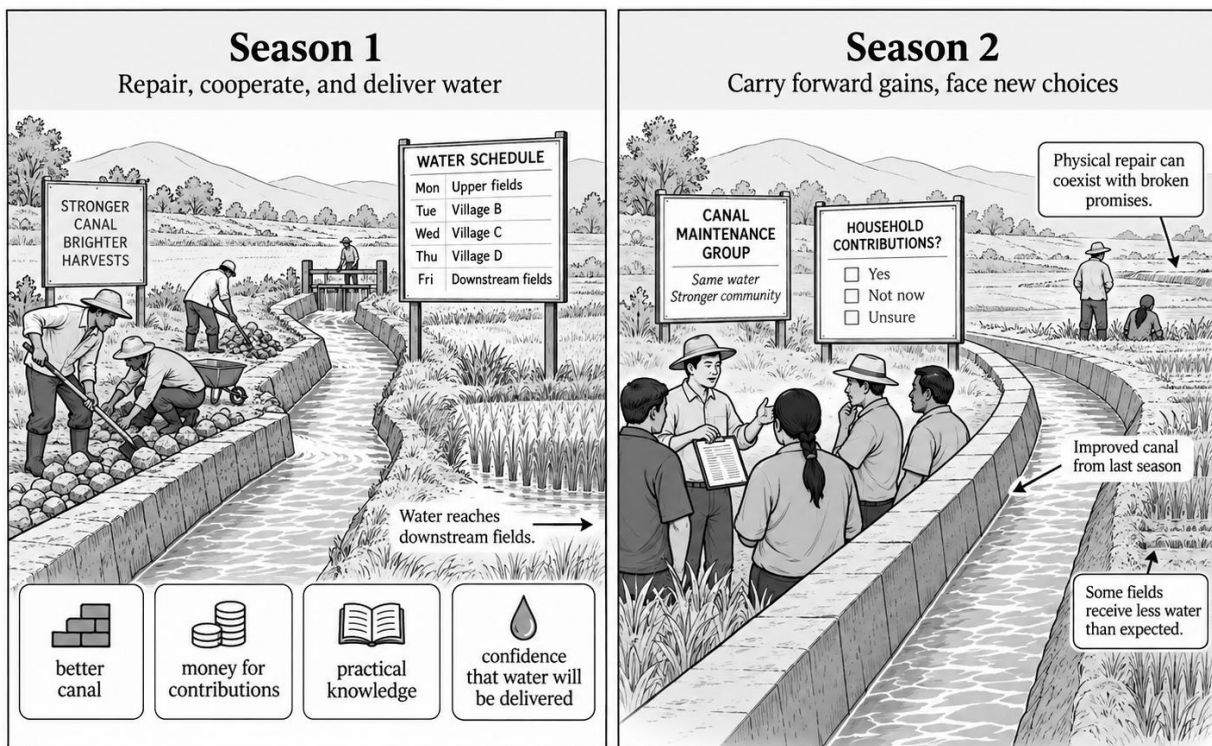
In our community, farmers agree to repair the canal and follow a water schedule. They complete the work, respect the schedule, and deliver water to downstream households during a dry period. The next season may then begin with a better canal, more money available for contributions, and greater confidence that a water claim will be honored. The organizers may also have learned how long repairs take and whom to involve.

Each consequence works differently. Money finances a contribution. Confidence affects willingness to contribute. Practical knowledge helps people use their contributions effectively. Someone can also use a tool purchased from earlier proceeds without knowing how the money was earned. Earlier outcomes enter later activity through physical and financial resources as well as through memory and judgment.

A completed repair can coexist with a broken promise. If water fails to reach the downstream fields, the canal's physical condition may improve while some farmers become less willing to maintain it. Another household's larger harvest could increase its ability to contribute. The next collective decision brings together participants whose resources and expectations have developed differently.

Feedback occurs when the consequences of an activity return to affect its later operation. Here, water allocation affects future maintenance participation, which affects subsequent water delivery. **The conditions of the second season are partly produced during the first.**

Figure 3. What one season leaves for the next



In the imagined community, repairs, income, and experience carry into the next season. A stronger canal can coexist with uneven water delivery and different willingness to contribute.

6.2 What makes cooperation worth repeating?

The comparative Nepal research gives us a reason to examine how farmers organize these relationships. Ostrom, Lam, and Lee reported that farmer-managed systems tended to achieve higher average performance than state-operated systems, despite often using simpler infrastructure. The finding challenges the presumption that more sophisticated engineering and centralized administration necessarily secure better irrigation performance. [45]

Bastakoti and Shivakoti found stronger enforcement in the farmer-managed systems they examined, alongside more favorable conditions for collective action and performance. Their analysis distinguishes rules developed by users from rules largely formulated by external officials. Self-governance here involves making and applying rules for shared activity. [46]

The upstream and downstream positions help explain why rules matter. An upstream farmer might obtain water while contributing little maintenance work. A downstream farmer might contribute substantial work yet receive unreliable water. Both depend on a functioning canal, but their immediate opportunities and obligations differ.

In our community, linking a recognized water claim to a maintenance obligation could make those obligations more dependable. Contribution records, checks on water delivery, and a way to raise disputes would let participants assess whether others had done their share and whether a complaint received a

response. A household that repeatedly works but loses its water turn has grounds for questioning the arrangement even while the written schedule remains in place.

Receiving promised water may sustain the participation needed to deliver water next season. Repeatedly losing it may weaken that participation. Confidence in the arrangement has concrete grounds to investigate: observed contributions, delivered water, and workable responses to complaints. These connect the allocation of benefits to the maintenance of capacity.

6.3 Experience has to be interpreted—and carried forward

Levitt and March describe organizational learning as the preservation of lessons from experience in routines that guide subsequent behavior. Such routines can transmit practical knowledge across changes in personnel, but they can also carry incomplete or mistaken inferences. [47]

A revised water schedule illustrates the problem of interpretation. A better harvest after its introduction might reflect the schedule, more rainfall, fewer unrelated crop losses, or a combination. Farmers still have to decide whether to retain or revise the schedule while some of those causes remain uncertain. Their interpretation can affect the next season's practice even when it is mistaken.

A lesson might become a pre-season inspection, a revised contribution record, or an established way of hearing complaints. A new organizer can follow the practice without having witnessed the original failure. The experience is carried forward through a routine as well as through the memories of those involved.

Ostrom and Gardner describe a concrete organizational response in the Kerabari irrigation system. Farmers in one branch considered those in the other insufficiently active in maintenance and operation. Users replaced a single committee with separate branch committees while retaining a common chairperson who owned land in both areas. [44, p. 108] The episode documents a response to dissatisfaction, rather than an estimate of the committee change's separate effect on performance.

The users revised part of the organization through which cooperation occurred. Turning dissatisfaction into such a decision requires both a way to get the complaint heard and authority to act on it.

6.4 Who can turn an outcome into influence?

Separate complaints about unreliable water can become a coordinated demand through representation, records of missed turns, and negotiation. These activities could give the downstream farmers a stronger voice in the committee. An upstream farmer might instead hold more land, control an important intake, or have closer relationships with committee members. Resources, physical access, and organizational relationships provide different possible routes to influence.

Here, *power* concerns the ability to shape relevant decisions and practices. Explaining its exercise requires following how resources, authority, relationships, or coordinated action affect a decision. An outcome matters to that explanation when it changes the ability to act in the next round.

Successful cooperation could leave our farmers with an association, reliable records, and negotiating experience that help them challenge a later proposal. Repeated failure could instead disperse participants

and weaken their ability to hold a committee accountable. The organizational result can extend beyond the water delivered in one season.

Acemoglu and Robinson model another route from resources to influence. In their account, a group losing formal political power may invest in other means of influence and preserve favorable economic institutions. Their model shows how formal authority and effective influence can develop differently under specified conditions. [48]

Pierson's account of policy feedback follows how policies reshape subsequent politics through resources and incentives, and through information that affects how people understand their position. A policy's operation can alter the setting of later political decisions. [49]

A program that helps establish user associations could therefore face a different political setting when its continuation is considered several years later. Newly organized groups might support it, seek different terms, or oppose its administration. They might also struggle to coordinate. An outcome can create an interest while the organization and expression of that interest determine whether it becomes an effective voice.

6.5 Local experience can change wider institutions

Chapter 2 followed the Household Responsibility System's development through experimentation, local protection, and wider acceptance. Here the relevant connection runs from a reported result to people with authority to permit further change. Deakin and Meng describe how local and provincial support helped the Xiaogang arrangement survive, while reports and political interpretation brought it wider attention. [14]

A report made an earlier outcome available to a new audience. Officials could use it in a policy argument; wider recognition could then alter the risks facing others considering a similar practice. In this reading of the historical account, local initiative gained wider reach through protection, interpretation, and public authority. [14]

The productivity research addressed a different question. McMillan, Whalley, and Zhu examined the contributions of price changes and changed incentives, attributing a substantial share of the measured increase to payment-system changes. [12] That attribution concerns what produced the improvement. The historical account concerns how reported experience helped secure support for reform.

Evidence that apparent success influenced a decision can be stronger than evidence that the practice caused all the improvement attributed to it. Evidence of a beneficial effect likewise leaves the timing and process of adoption to be explained. Persuasion and effectiveness have distinct evidentiary histories.

The connection to the canal is the work that turns experience into a rule revision. People notice and interpret an outcome, communicate it, and act through a body able to change what is permitted or required. Those steps give a local result consequences beyond its original participants.

6.6 Feedback can alter the bargain that sustained it

A more durable water-diversion structure could improve the canal while changing the bargain that secured its maintenance. Before the improvement, upstream farmers in our example depend on substantial

downstream labor to rebuild the structure. Afterward, much less labor is needed. Reduced dependence could weaken the upstream farmers' incentive to honor the agreement through which they obtained those contributions.

Lam's analysis of technological investment and irrigation governance in Nepal develops this concern: technical changes can intensify asymmetries among users and weaken incentives to contribute to maintenance. Engineering improvements alter the dependencies within which existing rules operate. [50]

Revising water obligations and maintenance contributions together could address the altered bargain. Preserving the same formal rules after a technical improvement may leave the participants with a different relationship from the one those rules originally supported.

A failed agreement can also prompt different responses. Downstream farmers might withdraw from maintenance after repeatedly missing their water turns. Worse delivery could encourage further withdrawal, reinforcing decline. Monitoring, negotiation, or replacement of an ineffective organizer could instead interrupt that sequence. Available resources, interpretations, relationships, and authority help explain which response becomes possible.

Building capacity can change the relationships through which it is maintained. A successful intervention may therefore require subsequent adaptation. Following feedback means identifying what the result altered and how that alteration sustains, weakens, or redirects the next round.

6.7 How do we know that an outcome changed what followed?

A community cooperates, the canal functions, and cooperation continues. To attribute that continuation to the earlier result, we need to distinguish feedback from other sources of persistence. Effective rules might support both seasons; water might remain abundant; or households more able to cooperate might stay while others leave.

An explanation through changed expectations requires evidence about what farmers believed before and after the first season, how those beliefs affected their decisions, and whether contributions changed. Interviews about confidence and records of attendance or repairs answer different parts of that inquiry.

A material explanation follows resources. If a larger harvest enabled later maintenance payments, the link concerns household funds and subsequent contributions. A rule-revision explanation follows the complaint, the decision process, and the practice that followed. Continued operation alone leaves these alternatives unresolved.

Lam and Ostrom studied irrigation assistance initiated in Nepal in the mid-1980s using observations from three periods. Their analysis followed combinations of infrastructure, organization, rules, sanctions, and collective action. They concluded that infrastructure investment alone was insufficient to sustain higher performance. [51] The longitudinal evidence connects an initial intervention with later performance; it is not a randomized estimate of one feedback mechanism.

A feedback explanation must identify the earlier outcome, what it changed, and how that change affected later activity. Observing those connections helps distinguish adaptation, continuing support from unchanged conditions, and persistence because participants cannot alter an arrangement they dislike.

6.8 How do advantages come to occur together?

Repeated irrigation activity can leave maintained equipment, funds for the next repair, expectations about others, and an organization able to act. These results can be unevenly shared. One household might finish a season with more resources and a stronger voice while another loses confidence in a system on which it still depends. Such histories help explain why the capacities, resources, and access observed together in Chapter 4 occur in those particular combinations.

The business in the Introduction gained income and a reference from an early contract. Equipment bought with the proceeds and experience gained through the work could then support the next order. Its current productive position was partly assembled through earlier outcomes. The other owner might improve a skill while losing savings, find another customer, or use shared equipment. Each history has to be followed through the particular resources and opportunities it develops or loses.

In the Dutch research-funding study, an initial award affected whether closely ranked applicants entered later competitions. Their subsequent funding histories diverged over eight years, even with the original award excluded from the comparison. The participation difference identifies a route from one allocation to later opportunities. [2]

Property can carry a result beyond the original participant. An income-producing asset retained by a business owner and transferred to a child gives the child a resource arising from activity in which the child did not participate. Rights of retention and transfer help determine that reach.

Associations and institutional reform add another route: earlier activity can change the rules for turning positions into returns. **Earlier outcomes can help produce both the positions in a later snapshot and the connections between their components.**

6.9 How can ordinary variation produce a skewed distribution?

The earlier examples combined changes in resources, cooperation, and rules. We can now keep the rules fixed and examine how repeated gains and losses change the resources carried into each new round. Begin with many participants who each hold 100 units of an asset, with equal ability and the same rules. Each round brings a proportional gain or loss, and the resulting holding becomes the starting amount for the next round. A 10 percent gain adds 10 units to a holding of 100 and 100 units to a holding of 1,000. The accumulated position determines the size of the next proportional gain or loss.

Over successive rounds, holdings change through a chain of multiplications. Logarithms let us express that multiplication as addition; taking the exponential reverses the conversion. This makes it possible to calculate how holdings are distributed after many rounds.

Several independent, normally distributed influences determine the logarithm of each round's growth factor. Taking the exponential produces a positive multiplier for the existing holding. Draws are independent across participants and rounds, and their distribution remains unchanged. Participants face the same chances but experience different sequences. The calculation note specifies the influences and their combination.

The growth factors are chosen so that each participant's expected holding remains 100. The realized total in a finite population can vary. Table 1 shows the distribution generated as the rounds accumulate.

Table 1. Holdings generated by an illustrative accumulation process

Completed rounds	Median holding	99th-percentile holding	99th percentile / median
0	100.0	100.0	1.00
1	98.0	156.1	1.59
10	81.9	356.6	4.35
50	36.8	987.4	26.84

Calculated population percentiles under the stated assumptions, not empirical observations or simulated sample results. Holdings are in hypothetical units; a round is not a specified calendar period. The 99th percentile is the level below which 99 percent of holdings fall, not the average holding of the top 1 percent.

After fifty rounds, the 99th-percentile holding is 26.84 times the median. **This is a mathematical illustration, not a fitted estimate of any real population.** The median marks the middle outcome; the mean also reflects the size of the largest holdings. Relatively uncommon sequences producing very large holdings keep this distribution's mean at 100 even as its median falls. The distribution of each round's influences is unchanged; their repeated operation on accumulated positions has widened the outcome distribution.

After any positive number of rounds, the logarithms of holdings are normally distributed and holdings follow a *lognormal distribution*, which is right-skewed with a long upper tail. [52] The illustration produces finite-time lognormal distributions, not a Pareto power-law tail. Research on wealth distributions examines other processes that can generate skewness and thick upper tails. [1]

Widening comes from differences in the proportional changes participants experience and the retention of each result for the next round. Identical proportional changes for everyone would preserve their holding ratios. Different sequences also allow someone ahead to fall behind, so increasing dispersion can coexist with movement among positions.

The normally distributed inputs represent repeated influences. Holding ability equal isolates the contribution of accumulated variation: each round acts on the position left by the previous one. This operation connects ordinary inputs to a markedly different cross-sectional distribution.

6.10 What changes when the process changes?

After each round, let participants place one-fifth of their holdings in a common pool and divide it equally. At that step, total holdings remain unchanged and every gap between two participants becomes four-fifths of its previous size. The next round's proportional changes act on these altered holdings.

Pooling changes what participants carry forward. New draws can widen their holdings again, while each pooling step contracts the gaps then present. The calculation establishes that stepwise contraction; the modified process's long-run distribution remains open.

Other arrangements act on different parts of a trajectory. Shared equipment could expand access to productive activity. A rule for dividing a firm's proceeds could alter what each participant retains. Support during an interruption could preserve a chance to participate again. Each modifies a particular connection between present activity and later possibilities.

The baseline accumulation example permits individual retention of gains. Pooling modifies that retention rule while leaving the specified growth influences unchanged. The comparison is between two arrangements, with their feasibility and value still to be assessed.

A lower 99th-percentile-to-median ratio could also result from less variable growth influences or fewer elapsed rounds. Inferring intervention from a narrower distribution therefore requires evidence about the transitions and the relevant alternative arrangement. The distribution alone leaves these explanations unresolved.

Arrangements differ in how they let gains and losses enter the next period. Pooling changes the holdings carried forward. A revised canal agreement changes what people contribute and receive. Each directs attention to a specific way an outcome can affect subsequent activity.

6.11 What does the next snapshot record?

The point-in-time view examines how current capacities, resources, relationships, and rules operate together to produce outcomes. The dynamic view follows how earlier activity formed those positions and how current outcomes alter what follows. A later snapshot records positions reached through that continuing history.

The accumulation example isolates one route from ordinary inputs to wide outcome differences. The wider discussion adds access, scale, claims, and the development of capabilities themselves. Evidence about these connections is needed to explain an actual distribution.

Mobility requires further information. Two populations could have the same four income levels every year, with the same people occupying each level in one population and people rotating through them in the other. Their annual distributions would be identical while their histories of persistence and mobility differed. Following people, entry, exit, and relationships reveals those differences.

The canal's next season may likewise begin with more reliable infrastructure and stronger cooperation, alongside unequal access or maintenance work concentrated on a few households. Understanding how these results are sustained brings their value into view. Chapter 7 asks **what counts as improvement—for whom, through which means, and over what period?**

Calculation note for Table 1

This note specifies the illustration; it is not an empirical model fitted to a population. Let participant i begin with $W_{i,0} = 100$. In each round, four independent standard-normal influences $\varepsilon_{ik,t}$, independent also across participants and rounds, determine a log growth factor:

$$Z_{i,t} = 0.1 \sum_{k=1}^4 \varepsilon_{ik,t} - 0.02, \quad W_{i,t+1} = W_{i,t} \exp(Z_{i,t}).$$

Thus $Z_{i,t}$ has mean -0.02 and variance 0.04 . The centering makes $\mathbb{E}[\exp(Z_{i,t})] = 1$, so $\mathbb{E}[W_{i,T}] = 100$. It does not hold the realized total of a finite population fixed. The illustration leaves consumption, debt, and entry or exit outside the calculation.

After T rounds, the natural logarithm of holdings is normal with mean $\ln(100) - 0.02T$ and variance $0.04T$. Its implied holding at population quantile p is

$$Q_T(p) = 100 \exp[-0.02T + 0.2\sqrt{T} \Phi^{-1}(p)],$$

where $\Phi^{-1}(p)$ is the corresponding standard-normal quantile. Table 1 uses $p = 0.50$ and $p = 0.99$, with $\Phi^{-1}(0.99) \approx 2.326347874$. These are exact distributional calculations, rounded for display, not a simulation. The rows describe finite-time distributions, not a stationary distribution. The lognormal formulas are documented in [52].

For the modification in Section 6.10, let $\overline{W}$ be the actual mean immediately before pooling. Each holding becomes $W_i' = 0.8W_i + 0.2\overline{W}$. Hence $W_i' - W_j' = 0.8(W_i - W_j)$, while the total is preserved. This verifies the stated contraction at the pooling step; it does not calculate the modified process's long-run distribution.

CHAPTER 7

What Counts as Improvement?

An irrigation system can sustain production while distributing water, maintenance work, and influence unevenly. The previous chapter followed how those results enter the next season. Evaluating the arrangement means asking which gains to preserve and which burdens or exclusions to change.

A society could become more productive, healthier, less equal, more politically free, and less able to sustain its use of natural resources over the same decade. Judging that trajectory requires attention to what improved, for whom, at whose expense, and with what consequences for the future. Relieving a serious deprivation can be an identifiable achievement even while other problems remain.

This chapter examines distinct evaluative perspectives rather than constructing one unified evaluative framework. Their differences help us identify the reasons for recognizing a gain, questioning a burden, or preferring another course of action.

7.1 What improved—and compared with what?

In the Oregon experiment, Medicaid coverage increased service use, reduced financial strain, and reduced positive screenings for depression over approximately two years. The study did not find statistically significant improvements in the selected physical-health measures. [18]

Financial protection and the measured mental-health gain are achievements to recognize. Assessing the combination begins with those specific gains and the reasons they matter. A verdict on the whole program also requires attention to its other consequences and costs.

The comparison can change what the judgment concerns. A community's health could deteriorate during an economic disruption while an intervention prevents an even larger decline. Health would then be worse than before but better than it would have been without the intervention. Comparing periods and evaluating the intervention answer different questions.

Another comparison concerns feasible alternatives. An intervention can help while another use of the resources would help more. In a local budget, financing a clinic might require postponing a water repair. The benefits of the clinic and the consequences of delayed repairs both enter the choice.

An identified gain is a reason for a favorable judgment. A judgment about the whole choice requires examining what else changes. A particular study can establish one contribution to that broader decision.

7.2 Who gains—and is compensation real?

Chapter 4's French manufacturing study located gains within robot-adopting firms alongside employment losses elsewhere in the market. [29] Evaluating the change requires including the people losing opportunities as well as those receiving gains.

Take two households with annual incomes of 20 and 80, holding household size and prices fixed and considering income alone. After a change, their incomes become 15 and 105. Combined income rises from 100 to 120, while the poorer household loses one-quarter of its income. Higher combined income coexists with a loss whose significance must be assessed in that household's life.

The winner's gain is large enough to compensate the loser in this example. A costless transfer of 6 would leave incomes of 21 and 99, above both starting incomes. Kaldor proposed judging a change by whether those gaining could compensate those losing while retaining a gain. This separates a test of potential aggregate benefit from a judgment about the justice of the resulting distribution. [53]

Actual compensation changes the outcome received. Before the transfer, the poorer household has 15; afterward, it has 21. The possibility of payment leaves the first position in place until someone makes the payment. An assessment that invokes compensation must therefore distinguish the resources available to provide it from the arrangements that deliver it.

Other losses raise further questions about adequacy. A lost right, an unwanted relocation, or an interrupted education requires an assessment of whether compensation is possible and what could make it sufficient. Money may not make the affected person whole.

The practical choice includes how the expansion is organized, whether avoidable losses can be prevented, and how promised compensation reaches those bearing the costs. Distribution is part of the change being evaluated.

7.3 What can people actually do?

Income and services matter partly through the lives they make possible. Sen's question, "Equality of What?", directs attention to what a person is able to do and to how differences among people and their circumstances affect the use of resources. Equal quantities of resources can offer different practical opportunities. [54]

The treatment example makes the difference visible. Two people hold the same entitlement at the same clinic. One can travel by bus and arrange time away from work; the other requires accessible transport that is unavailable. The same coverage leaves them with unequal opportunities to receive treatment.

The capability approach uses *capability* to mean a real opportunity to be or do something, rather than a personal skill alone. An achieved state or activity is called a *functioning*. Robeyns explains why an evaluation can concern both what people achieve and the meaningful alternatives available to them. [55]

Declining a training course illustrates the distinction. One person may prefer another available activity. Another may want to enroll but be unable to arrange childcare. The same enrollment record can reflect a choice among usable alternatives or the absence of an alternative the person wants.

Needs, abilities, commitments, and circumstances help determine whether an allocated benefit becomes usable. Individuality matters to that conversion: what supports one person's participation may leave another person's obstacle in place.

The distinction also connects to Chapter 4. A health service's capacity concerns its ability to provide treatment; a person's capability concerns the opportunity to obtain it. The service may become more productive while remaining inaccessible to particular patients. Removing a transport barrier could expand

a patient's opportunity while leaving clinical capacity unchanged. Provision and the person's usable opportunity require attention to different connections.

There is also a judgment about which opportunities deserve support. A greater ability to intimidate others is an expanded ability with objectionable consequences. The capability approach requires attention to the value of the opportunities being assessed. [55]

An account of improvement should ask whether people can actually use the possibilities whose expansion is being celebrated.

7.4 Do narrower gaps mean better lives?

The gap between the two households could narrow solely because the better-off household loses income, with no benefit to anyone else. Equality of income would increase while the poorer household's income remained unchanged. Whether the narrower gap counts as an improvement depends on the reasons for valuing it.

Parfit distinguishes concern with equality from giving priority to people because they are worse off. The latter supports helping someone in a bad position without valuing a reduction in someone else's advantage for its own sake. People can care about equality and other values together; the distinction identifies different reasons for judgment. [56]

In the household example, we can ask whether the change reduces deprivation, expands opportunity, or alters an objectionable relationship. Those questions specify what a smaller gap might mean for the people involved.

A different judgment arises when unequal rewards are defended through effort. Recall the workers whose access to retraining differed in Chapter 4. Completing the course required effort; being unable to afford time away from earnings constrained the other worker's opportunity to supply it. Their eventual earnings record the consequences of both effort and access.

Even an estimate isolating effort's contribution would leave the appropriate reward to be judged. Roemer's examination of equality of opportunity identifies the need for additional ethical judgments about rewards to effort. Explaining the contribution and deciding how much inequality it justifies are separate tasks. [57]

Luck raises a similar issue. An unchosen event can help explain a gain or loss. Deciding whether to insure against it, compensate its consequences, or leave the outcome in place requires reasons about responsibility, solidarity, rights, and acceptable risk. Personal action remains consequential within the circumstances in which it occurs.

Unequal outcomes do not, by themselves, establish unequal effort, unequal responsibility, or a justified scale of reward.

7.5 How was the result obtained?

The imagined canal could deliver the same quantity of water under two arrangements. In one, maintenance duties are negotiated, contributions are recorded, and farmers can challenge a missed turn. In the other, an organizer compels work through intimidation and excludes particular households from

raising complaints. Equal output coexists with a substantial difference in freedom, treatment, and participation.

Rights can constrain the means used to obtain a gain, even where violating them would increase output. Rawls gives an influential statement of this position by assigning basic liberties priority over economic and social gains. [58]

Anderson's account of democratic equality emphasizes relationships among people who can participate as equals. Subordination can make an arrangement objectionable alongside concerns about the goods people receive. [59] In the canal, a protected right to complain can matter both as a way to improve water delivery and as protection against dependence on an organizer's favor.

Effective voice requires more than an invitation to attend a meeting. A farmer who fears retaliation for speaking has limited scope to use it. Majority support also leaves the justice of imposed burdens open to examination. How a decision is authorized and what it permits are distinct concerns.

The agricultural productivity gain attributed to changed household incentives in the China case supplies another achievement to recognize. [12] Security of claims and the ability to challenge a decision require additional examination; the productivity study does not establish harms on those dimensions. Together, the questions concern what a person receives, what the person can decide, what others may demand, and what recourse exists when a claim is denied.

Institutions matter to evaluation through both their effects and the relationships they establish among people. Those relationships can themselves be part of what makes a social arrangement better or worse.

7.6 Whose future enters the judgment?

Chapter 5 distinguished several turning points in recovery. The U.S. recession ended in June 2009, while payroll employment continued falling until February 2010. [41] The endpoint chosen for an assessment determines which part of that experience is visible.

A household regaining its former income after several years has reached one recovery benchmark. The intervening losses remain part of its history, along with any effects on savings, health, or opportunity. Evaluating the path requires keeping those costs visible alongside the later position. A later benefit may outweigh an earlier cost under a defensible judgment, but that judgment must include both.

Future provision matters in the same way. The imagined community could raise this year's harvest by postponing essential maintenance or drawing down a water reserve that cannot be replenished. Part of the current gain would then come from weakening the conditions of later production.

Dasgupta's account of national wealth evaluates the productive assets, including natural assets, that support well-being over time. Fleurbaey and Blanchet likewise distinguish assessing current welfare from assessing whether it can be sustained. [60, 61] For the canal, this means examining the infrastructure and water supply left for subsequent users alongside the value of the present crop.

A forecast can clarify the likely consequences of postponing repairs. Judging whether the current farmers' additional income justifies risks to future farmers also requires reasons about need, responsibility, available alternatives, and obligations across time. Later users' interests must enter that judgment explicitly.

Uncertainty brings the quality of the decision into view. A reasonable investment can fail, while a poorly justified gamble can succeed. Evaluation can examine the information available when the decision was made, the understanding of risks, the voice of those bearing them, and the availability of less harmful alternatives. These considerations distinguish a defensible decision from a fortunate result.

7.7 Do we need one score?

Income, opportunity, rights, distribution, and future interests can all matter to a decision. A summary measure can make patterns visible, while a decision model can state priorities and show how a conclusion changes when those priorities change. Their usefulness depends partly on making the judgments behind the numbers open to examination.

Fleurbaey and Blanchet's discussion of alternatives to GDP emphasizes the substantive choices involved in measuring welfare and sustainability. Selecting a measure requires deciding what it should represent. A larger collection of indicators still leaves their roles and relative importance to be judged. [61]

Two irrigation proposals illustrate the task. One produces a larger total harvest. The other produces a smaller total increase but substantially improves water reliability for households with the least secure access. A combined score must specify the weight given to reliability relative to output and whether the beneficiaries' initial positions affect that weight. Those choices need reasons that readers and affected people can examine.

Some comparisons allow a clear preference without settling every priority: one option can improve an important condition without worsening another that matters. Other comparisons depend on resolving conflicts among gains and burdens. A judgment remains open to criticism when it ignores a serious loss, applies inconsistent standards, excludes affected people without justification, or rests on a false account of the consequences.

Disagreement about what a proposal will do calls for better evidence. Agreement on its likely consequences can still leave disagreement about which deserve priority. Further measurement can expose mistaken premises, while the priority itself requires evaluative reasons.

Practical choice can proceed by eliminating clearly inferior options, protecting essential rights, and identifying a defensible preference among the feasible alternatives. Revising a proposal may also reduce the conflict that made the original comparison difficult. A complete ranking of every possible social arrangement is unnecessary for that task.

Preserving the present arrangement belongs in the comparison too. Postponement could leave unreliable water access and unfair maintenance duties in place. Inaction has consequences for the people who continue to live under those conditions.

7.8 What can we responsibly call improvement?

Return to the canal with a repair and rule change that gives downstream households more reliable water, preserves adequate access upstream, reduces total repair work, and provides an effective way to challenge allocation decisions. Assume its costs are affordable, maintenance is funded, and the evidence supports the durability of these gains.

On those criteria, there is a strong case for improvement. Valuable opportunities expand, a serious disadvantage is reduced, cooperation requires less work, and future provision is sustained. The judgment concerns those achievements; other problems can remain open to criticism and further change.

A larger harvest obtained by silencing one group and imposing an unreasonable unpaid burden on another would call for a different overall judgment. The gain in output would remain, alongside losses that the increased total could not make disappear.

The point-in-time and dynamic views help specify the subjects of evaluation. We can examine the positions people occupy now and the opportunities they can use, then follow the burdens experienced along the way and the conditions left for later participants. Both the endpoint and the path enter the judgment.

Understanding what happened is necessary for evaluation, but it does not complete evaluation. We must explain why particular gains matter, which burdens are acceptable, whose interests count, and what responsibilities extend beyond the present.

Several judgments can be defensible while remaining answerable to evidence and reasons. We can recognize a gain, reject an unjust means, question a neglected loss, and seek a better alternative. The standards concern the lives, relationships, and possibilities a trajectory changes.

Conclusion

The two businesses in the Introduction can now be understood through both their present operations and their earlier experience. Current earnings depend on usable equipment, customer access, working relationships, and claims on the proceeds. Earlier contracts, retained income, and decisions help explain how those resources and relationships developed. The point-in-time view follows how the present position operates; the dynamic view follows how it was formed and what it makes possible next.

We began by asking why relatively modest differences in individual capabilities can coexist with much larger differences in rewards, resources, and influence. **The answer lies in how differences are combined and made consequential, and in how those consequences enter later activity.** A modest performance difference can receive a large reward, and that reward can affect several later opportunities. Its scale and continuing effects require explanation alongside the contribution of the person receiving it.

How present conditions become outcomes

Skills become usable through equipment, time, finance, and access. Knowing the separate distribution of each component leaves their combinations to be examined. Advantages may accompany one another, or support may compensate for a disadvantage. Chapter 4's pairing illustration showed how the same component distributions and outcome rule can produce different distributions when the pairings change.

Organization makes those combinations productive. A suitable operator, a functioning machine, and a workable production schedule allow an order to be completed. A shared facility can provide access to equipment for someone unable to buy it. What a person brings into an activity includes resources made available through relationships as well as personal holdings.

Selection, scale, and claims then shape the return. An exclusive order can give one of two closely matched suppliers all the revenue from that order. The size of the order and the organization able to fulfill it affect how large that difference becomes. Ownership and payment arrangements determine who receives the proceeds of the joint activity.

The music-market experiment made the social process visible in another setting. Stronger social influence increased inequality and unpredictability of success among the same set of songs. Appeal remained relevant, while information about previous choices affected the distribution of success. [3]

People contribute through the work they can do and the decisions they make. The business owner's skill may be essential to completing an order. Explaining the owner's reward also requires the order's value, the division of proceeds, and the access that made the work possible. These connections allow rewards to differ far more than performance, or allow shared resources and benefits to moderate the difference.

The point-in-time account brings those connections into view under the conditions operating then. Following their formation takes the explanation into the history of the people, resources, and rules involved.

How outcomes change later conditions

The people entering an activity already have histories. The neighborhood evidence linked childhood circumstances and the timing of a move to later outcomes. [5] The development of abilities and opportunities belongs in the explanation alongside the choices people make with them.

Resources and relationships develop through activity too. In the business example, an early contract produced income and a customer reference. Retained proceeds financed equipment; experience and recognition affected what could be attempted next. Several advantages observed together were partly produced together.

The Dutch funding study supplied evidence of this connection: an early award affected participation in subsequent competitions and later funding among closely ranked applicants. [2] The allocation helped shape the conditions under which later opportunities were pursued and awarded.

Property extends some consequences across activities and generations. A retained asset can supply returns or a reserve; transferring it can alter another person's starting position. The rights attached to the asset determine which of these routes is available. The present holder's resources can thus include the proceeds of activity performed by someone else.

Reserves also affect what an unchosen event does to a trajectory. Waiting for a customer is manageable while the business can finance continued operation. The same wait can end in closure when cash runs out. Timing may be outside the owner's control, while its consequences depend on accumulated resources and current obligations. Luck enters through that encounter between an event and a particular position.

Earlier results can also support efforts to revise the rules. An association can use experience and resources gained through cooperation to seek different terms. A response to exclusion can broaden access; another response can preserve an advantage. The authority and organization through which these efforts succeed become part of the explanation.

Dynamics therefore includes the development or erosion of capabilities, the retention and transfer of resources, the formation of relationships, and the revision of rules. Earlier outcomes can reinforce a position, support recovery, or redirect subsequent activity. Each later snapshot records a stage of these developments.

Reading a distribution through its history

Chapter 6 isolated one route from ordinary influences to a widely dispersed cross-section. Starting with equal holdings and the same chances, participants experienced different sequences of proportional changes. Repetition on accumulated holdings produced a right-skewed lognormal distribution from normally distributed log-growth influences. [52] The finite-time calculation demonstrates a possibility under its stated assumptions; it is not a fitted population estimate or a derivation of a Pareto tail.

The wider discussion identifies other connections to investigate: combinations, access, selection, scale, claims, and opportunities to continue after a setback. Evidence about their operation determines which explain a particular distribution. A snapshot locates positions; following people over time reveals who remains, moves, enters, or leaves.

A narrower distribution could reflect weaker amplification, a counteracting arrangement, or a different elapsed history. Distinguishing those possibilities requires evidence about transitions and the relevant comparison. Distributional shape alone does not identify the amount or effect of intervention.

Table 2 gathers the connections developed across the chapters. It points back to the cases and illustrations through which each can be followed.

Table 2. How differences become outcomes—and outcomes become later conditions

Connection	What it changes	Where the discussion develops it
Formation through experience	The abilities, expectations, and opportunities people bring into later activity.	Childhood neighborhoods and access to a move: Chapter 1, §§ 1.2–1.6.
Pairing, access, and organization	Which skills and resources become usable together, and who can realize a claim or productive possibility.	Medicaid: §§ 3.3–3.6. Preparation/support pairing and productive combinations: § 4.7.
Selection, scale, and claims	The size and division of returns, including rewards from activity involving many people.	Household production rights: §§ 2.1–2.3. Firms and contracts: §§ 4.3, 4.7.
Retention, accumulation, and transfer	Earlier proceeds become assets, reserves, or another person's starting resources; losses reduce what remains available.	Property: § 2.6. Business, asset transfer, and accumulation examples: §§ 6.8–6.10.
Timing, exposure, and response	Whether a disturbance can be absorbed, and whether assistance preserves an operating activity or arrives after it has broken up.	Financial crisis and the manufacturer: §§ 5.1–5.6.
Learning, cooperation, and maintenance	Experience affects expectations, routines, and participation that sustain or weaken provision.	Nepal irrigation and organizational learning: §§ 6.1–6.3, 6.6–6.7.
Influence and rule revision	Resources, organization, and reports of results can affect who can be heard and which rules govern later activity.	User associations and Chinese reform: §§ 6.4–6.5.

These connections can operate together. A retained gain can finance training, support another attempt, or help organize a demand for different rules. Following such links explains how the outcome of one activity becomes a resource, opportunity, or constraint in another.

Which possibilities should change?

An explanation becomes practically useful when it locates a connection people can act on. Shared equipment could broaden access; a contract could alter the division of proceeds; support during an interruption could preserve a chance to continue. Each modifies a particular part of the process. The relevant comparison is between specified arrangements, including the rules about access, control, and retention already in place.

The Nepal irrigation discussion brings continued provision into this choice. Lam and Ostrom's longitudinal study found that infrastructure investment alone was insufficient to sustain higher

performance; organization, rules, and collective action mattered to what followed. [51] The practical concern is both who receives water and how participation maintains the ability to deliver it. A change may need to develop capacity, maintain it, or make it accessible as well as alter the division of its benefits.

People can act within a limited reach. An owner can improve a product while customer demand remains uncertain; farmers can repair a canal while rainfall remains beyond their control; an agency can remove an identified barrier while later obstacles remain. The responsibilities of these participants also differ with their authority, knowledge, obligations, and available alternatives.

Judgment concerns both the gains and the means of obtaining them. A larger total can coexist with an unacceptable burden. A narrower gap can leave the worse-off person's position unchanged. A recovered income can follow years of losses. Evaluating the trajectory requires attention to usable opportunities, treatment and voice, burdens along the path, and the resources left for future participants.

The successful owner's effort and decisions remain contributions to recognize. Understanding the conditions that made them effective helps explain the achievement; deciding the appropriate reward requires reasons about what should be recognized and how. The scale of the observed reward is part of that inquiry, rather than its answer.

The opening puzzle has led to concerns people can recognize in their own lives: whether effort has a usable opportunity, whether a claim becomes a benefit, whether a setback can be survived, whether an objection can be heard, and whether an achievement can be sustained. The distribution matters because it directs attention to the relationships and histories behind those possibilities.

The outcome distribution records what individual differences and circumstances became through particular relationships, rules, and histories. Individuality, institutions, and luck become consequential together. Following their interaction helps us understand the positions we observe, how they enter what happens next, and which possibilities we have reason to preserve or change.

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